

BUS33 - 4

Labour Economics & Inequalities

[Module page](#)

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About this textbook

This year, for the first time, every Bachelor course at Albert School has its own textbook, thanks to your teachers and AI.

- The textbook is a help for working on the course at home, after the class.
- The course as taught in class remains the reference, and prevails over the textbook wherever the two differ.
- This is a first version. Help us improve it by sharing your feedback with your teacher.

Contents

Preface	5
1. Labour supply: the work–leisure choice	6
1.1. The constrained-optimisation set-up	6
1.2. Income and substitution effects of a wage change	6
1.3. Participation and the reservation wage	7
2. Labour demand: the marginal-product condition	10
2.1. Deriving demand from marginal product	10
2.2. Capital–labour substitution and the automation shock	11
3. Human capital	13
3.1. Becker’s model and the rate of return to education	13
3.2. Estimating the return on real data: the Mincer equation	14
3.3. Human capital versus signalling	15
4. Wage structure: differentials, discrimination, decomposition	17
4.1. Compensating differentials (Rosen)	17
4.2. Two kinds of discrimination	17
4.3. The Oaxaca–Blinder decomposition	18
5. Unemployment: frictions, matching, and the natural rate	20
5.1. Three kinds of unemployment	20
5.2. The Mortensen–Pissarides matching model	20
5.3. Reading the Beveridge curve	21
5.4. The NAIRU	22
6. Institutions and the minimum-wage debate	23
6.1. The competitive case against wage floors and insurance	23
6.2. Why these institutions exist all the same	24
6.3. The Card–Krueger reframing and monopsony	25
7. Inequality measurement	27
7.1. The Lorenz curve	27
7.2. The Gini coefficient	27
7.3. Percentile ratios and top shares	29
8. Wealth inequality and its dynamics	31
8.1. Two distinct statements: the aggregate ratio and the concentration mechanism	31
8.2. Intergenerational mobility and the Great Gatsby Curve	32
8.3. Is inequality really rising? Two objections	32
8.4. Why inequality matters: efficiency, not only fairness	33
9. Causal inference for labour-market policy	35
9.1. The counterfactual problem	35
9.2. Randomised controlled trials: manufacturing a counterfactual	35
9.3. Natural experiments: finding a counterfactual	36
9.4. Instrumental variables and regression discontinuity: building a clever counterfactual ..	36
9.5. Difference-in-differences: generalising the comparison	38
10. Automation and labour-market polarisation	40
10.1. Routine versus non-routine tasks	40
10.2. Polarisation and complementarity	40
10.3. Policy responses: active labour-market policies and flexicurity	41
Primary sources	43

Further reading 44

Preface

You arrive able to reason about a consumer at a tangency, a firm at $MC = MR$, and a market that clears — the microeconomics of your first semester — and about aggregate output, the Phillips curve, and the policy levers that move a whole economy — the macroeconomics of your second. This course aims that toolkit at one market in particular, the **labour market**, and by the end it will let you do a set of things those two courses did not equip you for. You will be able to:

- model a worker's labour supply as a work–leisure choice, split a wage change into its income and substitution effects, and say when the supply curve bends backward;
- derive a firm's labour demand from the marginal product of labour and predict how employment responds to a wage rise or a cheaper machine;
- compute the internal rate of return to a degree, and tell the **human-capital** reading of a wage premium from the **signalling** one;
- decompose an observed wage gap into an explained and an unexplained part, and say precisely what the unexplained part is and is not;
- explain why unemployment persists even with vacancies open, read a Beveridge curve, and separate frictional, structural, and cyclical unemployment;
- judge what a minimum wage, a union, or unemployment insurance does to employment — using the best evidence, not the textbook prediction alone;
- measure a distribution with a Lorenz curve, a Gini coefficient, and percentile ratios, follow Piketty's $r > g$ mechanism, read global inequality off the elephant curve, and argue why inequality carries **efficiency** costs and not only moral ones;
- and — the core habit this course builds — tell a **causal** claim from a **correlational** one, by naming the counterfactual a study constructs and the assumption that identifies it.

None of this is available to you now; all of it will be by the last page. Monetary figures are written in euros purely for concreteness; every argument is a statement about ratios and rates that reads the same in any currency.

1. Labour supply: the work–leisure choice

The supply of labour is not a primitive; it is the output of a choice every worker makes — how much of a fixed endowment of time to sell. Your microeconomics course taught you to read any choice as constrained optimisation. Here the good being “bought” with forgone wages is **leisure**, and its price is the **wage**.

1.1. The constrained-optimisation set-up

Example — the cost of an afternoon off. You earn €20 an hour. Taking Friday afternoon off — four hours — is not free: it costs you €80 of forgone pay. That €80 is the **price of leisure**. If your wage rose to €40, the same afternoon would cost €160: leisure has become twice as expensive, even though nothing about the afternoon changed. Every labour-supply decision is a purchase of leisure at a price equal to the wage.

Definition 1. A worker splits a fixed time endowment T between hours of work h and leisure $\ell = T - h$. With hourly wage w and non-labour income V (benefits, a partner’s income, capital income), consumption is $c = wh + V = w(T - \ell) + V$. The worker solves

$$\max_{c, \ell} u(c, \ell) \quad \text{s.t.} \quad c = w(T - \ell) + V.$$

At the optimum the marginal rate of substitution between leisure and consumption equals the real price of leisure:

$$(\text{MRS})_{\ell, c} = \frac{u_{\ell}}{u_c} = w.$$

The wage is the slope of the budget line and the opportunity cost of an hour not worked.

Read the condition as a stopping rule. If an extra hour of leisure is worth more to you than the wage it costs ($u_{\ell}/u_c > w$), you work less; if it is worth less, you work more; you stop adjusting when the two are equal. Everything in this chapter is a consequence of what happens to that balance when w moves.

1.2. Income and substitution effects of a wage change

This is the price-change decomposition from your microeconomics course, applied to the price of leisure — and it produces a result with no analogue in ordinary consumer theory.

Example — why a raise can make you work less. A nurse earning €25/hour is offered €35/hour. Two pulls act at once. Leisure is now dearer relative to consumption, tempting her to work **more** (substitution). But she is also richer at any given hours, and if leisure is a normal good she will want to consume more of it — to work **less** (income). Which pull wins is an empirical question, and for most workers the two roughly cancel or the substitution effect wins slightly, so measured hours change little when the wage rises. The income effect **can** nonetheless dominate — the clearest case

is the long fall in the average working week over the last century as real wages rose several-fold — and when it does, the supply curve **bends backward**: a higher wage buys fewer hours, not more.

Proposition 1. A rise in w decomposes into two opposing effects on desired hours:

- **Substitution effect**: leisure becomes more expensive relative to consumption, raising desired hours (always positive, for a wage rise).
- **Income effect**: the higher wage raises real income; if leisure is a normal good this raises desired leisure, lowering hours.

The net effect is **ambiguous**. When the income effect dominates, the individual labour-supply curve is **backward-bending**.

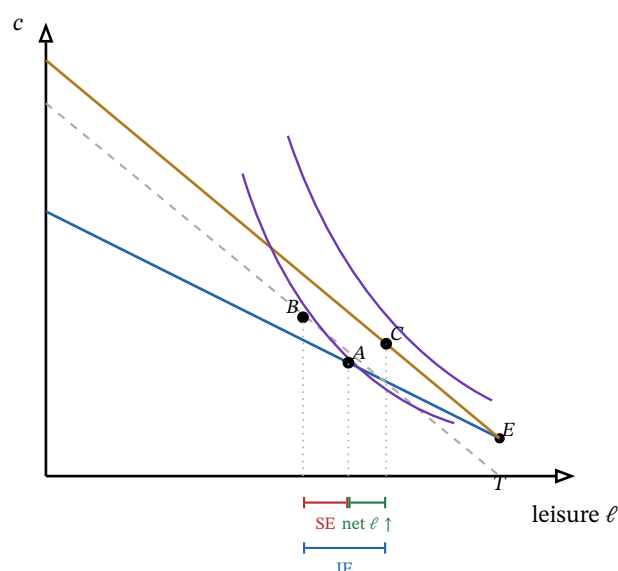


Figure 1: Decomposing a wage rise on the work-leisure diagram. The budget line pivots up around the endowment E (steeper = higher wage). The pure **substitution** effect $A \rightarrow B$ — slide along the original indifference curve to the new slope — cuts leisure. The **income** effect $B \rightarrow C$ — jump to the higher curve — raises it. The **net** change in leisure is the shorter span $A \rightarrow C$, the income effect less the substitution effect. Here the income effect wins: net leisure rises ($\ell_C > \ell_A$), so hours fall and supply bends backward.

Pitfall. The substitution effect of a **wage rise** always pushes toward more hours; only the income effect can reverse the sign. So a backward bend is evidence the income effect dominates — never the other way round. And note the asymmetry with ordinary goods: the worker is an **endowment** holder (of time), so a wage change rotates the budget line around E rather than shifting it, and the income effect is large precisely because labour income is most of total income.

1.3. Participation and the reservation wage

Everything so far is the **intensive** margin — **how many** hours to work, taking for granted that the worker works at all. But the first choice is the **extensive** margin: **whether** to work. It turns on a single threshold wage.

Example — the lowest wage worth accepting. A parent of a young child values an hour at home highly. Offered €8 an hour they stay home — the wage does not cover what that hour is worth to them. Offered €18 they take the job. Somewhere between lies the wage at which they are just indifferent between working and not: their **reservation wage**. Below it they supply **zero** hours; at it they enter the market.

Definition 2. The **reservation wage** w_R is the lowest wage at which a worker supplies any hours at all. Evaluate the choice at the no-work corner — all time taken as leisure ($\ell = T$), consumption equal to non-labour income V . There the reservation wage is the value of the **first** hour of work: the marginal rate of substitution between leisure and consumption at that point,

$$w_R = \frac{u_\ell(V, T)}{u_c(V, T)} = (\text{MRS})_{\ell, c} \text{ evaluated at } \ell = T.$$

If the market wage $w > w_R$, the budget line is steeper than the indifference curve at the endowment E , so moving into work reaches a higher curve: the worker **participates**. If $w \leq w_R$ they stay at the corner E and supply zero hours. A larger non-labour income V raises w_R (leisure is worth more when consumption is already high), making participation less likely.

Put the two margins together and the **individual labour-supply curve** is complete (Figure 2): flat on the axis at zero hours while $w < w_R$; then, once w crosses w_R , positive and at first **rising** with the wage (substitution dominates); and possibly, at high wages, **bending back** (income dominates, as in Figure 1). The reservation wage is where the curve **begins**; the backward bend is what it may do later.

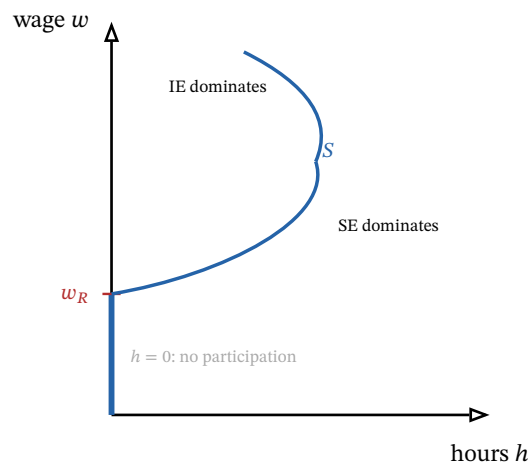


Figure 2: The individual labour-supply curve in (h, w) space. For $w < w_R$ the worker supplies zero hours (the segment on the axis) — the **participation** or extensive margin. At the **reservation wage** w_R the curve leaves the axis; above it, hours first rise as the wage climbs (substitution effect dominates), then can **bend back** at high wages when the income effect takes over — the same reversal built on the work–leisure diagram of Figure 1.

Exercises

1.1. A worker has time endowment T , wage w , and non-labour income V . (a) Write the budget constraint and state the tangency condition for optimal leisure. (b) Define the reservation wage, give its expression at the no-work corner, and show where it sits on the individual labour-

supply curve; state how a rise in non-labour income V changes the participation decision. (c) Decompose the effect of a wage rise into income and substitution effects, and explain, with the work–leisure diagram, the conditions under which the individual labour-supply curve bends backward.

Answers. (1) Budget: $c = w(T - \ell) + V$; tangency $u_\ell/u_c = w$. Reservation wage $w_R = u_\ell(V, T)/u_c(V, T)$, the MRS at the endowment: the worker supplies zero hours while $w < w_R$ and participates once $w > w_R$ — it is where the supply curve leaves the axis (Figure 2). A higher V raises w_R , making participation less likely. A wage rise raises desired hours through substitution (leisure dearer) but lowers them through the income effect (leisure normal, worker richer); the curve bends backward when the income effect dominates (Figure 1).

2. Labour demand: the marginal-product condition

The other blade of the market. A firm does not demand labour for its own sake but for what an extra worker **produces**: labour demand is a **derived demand**, derived from the demand for the output labour makes. This is the firm theory from your microeconomics course, re-read with labour as the variable input.

2.1. Deriving demand from marginal product

Example — the last barista. A café owner asks: should I hire one more barista? The new hire serves €120 of coffee a day (her marginal product valued at the menu price). She costs €100 in wages. Hire her — €20 of surplus. The owner keeps hiring until the **next** barista would add exactly €100, no more. Because each additional worker in a fixed café adds a little less than the last (diminishing returns), there is a unique stopping point, and it falls when wages rise.

Definition 3. A competitive firm with production function $Y = F(K, L)$ hires labour until the **value of the marginal product of labour** equals the wage. Write MP_L for the **marginal product of labour** — the extra output the L -th worker adds, capital held fixed. With output price p ,

$$\underbrace{p \cdot MP_L}_{VMP_L} = w, \quad \text{equivalently} \quad MP_L = \frac{w}{p}.$$

Because MP_L falls as L rises (diminishing returns), the VMP_L schedule slopes **downward**, and it **is** the firm's labour-demand curve: more labour is demanded only at a lower (real) wage.

When the firm also has market power in its **output** market, the object that matters is the marginal *revenue* product $MRP_L = MR \cdot MP_L$; under output-market competition price equals marginal revenue, so $MRP_L = VMP_L$. Monopsony in Chapter 6 brings MRP_L back.

Example — putting a number on the stopping point. Suppose $MP_L = 20 - 2L$ (cups per hour that the L -th worker adds) and each cup sells for $p = €6$, so $VMP_L = 6(20 - 2L) = 120 - 12L$ euros. At a wage of $w = €60$ the firm hires until $120 - 12L = 60$, i.e. $L = 5$ workers. Raise the wage to $w = €84$ and the condition $120 - 12L = 84$ gives $L = 3$: two jobs gone. The **size** of that fall — here 2 workers for a €24 wage rise — is what the next section makes precise.

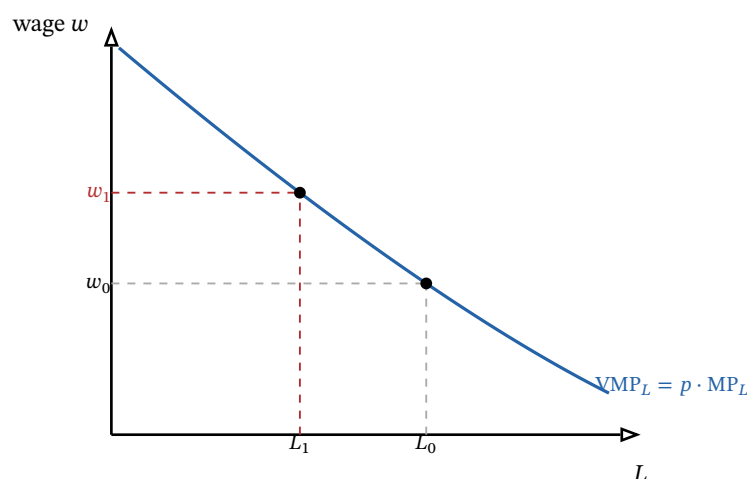


Figure 3: Labour demand is the value-of-marginal-product schedule. A wage rise from w_0 to w_1 moves the firm up the curve, cutting employment from L_0 to L_1 . How **much** employment falls depends on the curve's slope — the elasticity of labour demand — which in turn reflects how easily capital substitutes for labour.

2.2. Capital–labour substitution and the automation shock

How far the firm cuts labour when wages rise depends on whether it can swap in **capital**. That swap is governed by one elasticity.

Definition 4. The **elasticity of substitution** $\sigma \geq 0$ between capital and labour measures how the input mix responds to the relative factor price w/r (with r the rental price of capital). It is the ratio of the **percentage change** in the capital–labour ratio K/L to the **percentage change** in the relative price w/r that caused it:

$$\sigma = \frac{\text{percentage change in } K/L}{\text{percentage change in } w/r}.$$

A **larger** σ means capital substitutes more readily for labour: when w/r rises (a wage rise, or a labour-saving technology that cheapens effective capital) the firm re-mixes toward capital more aggressively, which makes its labour demand **more wage-elastic**. A **small** σ means the two are poor substitutes and the input mix barely responds.

Note carefully what σ measures: the response of the input **ratio** K/L , not of the employment **level** L — a distinction the next example turns on.

Example — reading sigma off a shock. Two plants each face a 10% rise in w/r (a labour-saving machine has cheapened capital). In plant H , $\sigma = 1.5$: the capital-to-labour ratio K/L jumps by $1.5 \times 10\% = 15\%$ — the firm re-mixes sharply toward machines. In plant L , $\sigma = 0.3$: K/L rises only 3% and the mix barely moves. Read this correctly: σ pins down the **input mix**, not the **headcount**. A 15% rise in K/L says capital grows 15% faster than labour — it does **not** say employment falls 15% (with capital pouring in, L falls by much less than that, and by how much depends on the plant's output too). What the comparison does show cleanly is that the plant with the larger σ re-mixes away from labour far more readily.

The employment **level** response has two pieces. A rise in w/r makes the firm **substitute** capital for labour at a given output (the substitution effect, stronger the larger σ), and — for a wage rise — it lifts marginal cost, so the firm **produces and hires less** (the scale effect). The **elasticity of labour demand**, the size of the employment fall, therefore rises with σ but never equals it: σ times the percentage change in w/r is the percentage change in the **ratio** K/L , of which only a fraction (capital's cost share) is the substitution-driven fall in L , with the scale effect added on top. For a **cheaper machine** the substitution piece still sheds labour, but the scale piece now runs the other way — lower costs expand output and can pull hiring back up — which is precisely why automation's net effect on employment is ambiguous.

Remark. A **labour-saving automation shock** enters exactly here. A cheaper, more capable machine lowers the effective price of capital r , raising w/r . If σ is large the firm substitutes machines for workers and labour demand contracts; if the technology instead raises MP_L (machines that make workers more productive), labour demand can **rise**. Whether automation destroys or creates jobs is, at root, a question about σ and about which margin the technology hits — the theme we return to in Chapter 10.

Exercises

2.1. A competitive bakery has $MP_L = 30 - 3L$ (loaves added by the L -th worker) and sells each loaf at $p = €5$. (a) Write the value-of-marginal-product schedule. (b) Find employment at a wage of €45 and at €75. (c) Explain why the schedule slopes downward and why the €30 wage rise costs the firm workers.

2.2. Two plants face a 6% rise in the relative factor price w/r after a labour-saving machine cheapens capital. Plant X has elasticity of substitution $\sigma = 1.5$; plant Y has $\sigma = 0.4$. (a) For each plant, compute the change in the capital-to-labour ratio K/L . (b) State why neither number is the percentage fall in employment. (c) Say which plant sheds proportionally more labour, and why.

Answers. (1) $VMP_L = 5(30 - 3L) = 150 - 15L$. At $w = 45$: $150 - 15L = 45 \rightarrow L = 7$; at $w = 75$: $150 - 15L = 75 \rightarrow L = 5$ — the €30 rise costs two jobs. Downward-sloping because MP_L falls with L (diminishing returns), so an extra worker is worth hiring only at a lower wage.

(2) The percentage change in K/L equals σ times the percentage change in w/r , so plant X: $1.5 \times 6\% = 9\%$; plant Y: $0.4 \times 6\% = 2.4\%$. Neither is the employment fall: σ governs the input **ratio** K/L , of which only a fraction (capital's cost share) is the substitution-driven drop in L , with the scale effect added on top. Plant X ($\sigma = 1.5$) re-mixes away from labour far more readily, so it sheds proportionally more.

3. Human capital

Workers are not interchangeable units of “labour”; they carry skills, and skills are **produced** by costly investment. Becker’s insight was to treat schooling and training with the same present-value logic a firm uses for a machine.

3.1. Becker’s model and the rate of return to education

Example — is the degree worth it? A student weighs a three-year degree. It costs €15 000 a year in fees and forgone wages — €45 000 in total. In return she expects to earn €6 000 more per year for a 40-year career. Is it worth it? Not by comparing €45 000 with $40 \times €6000 = €240000$ — euros arrive at different dates and must be discounted. The decision turns on the **rate of return** that equates the two present values. Let us actually compute it.

Before the rate of return, one idea your earlier courses did not cover: **discounting**. A euro promised a year from now is worth less than a euro in hand today — not because of inflation, but because a euro today could be lent out at the going interest rate and **become** more than a euro in a year. Run that logic backwards and a euro arriving later is worth today only the smaller sum that, invested now, would grow into it.

Definition 5. The **present value** of an amount x to be received t years from now, at discount rate r , is

$$PV = \frac{x}{(1+r)^t}.$$

Dividing by $(1+r)^t$ undoes t years of compound growth at rate r : it is the sum you would need today to reproduce x at date t . The present value of a whole stream of future amounts is the sum of the present values of its parts. A higher discount rate r , or a later date t , makes a future euro worth less today.

Definition 6. Becker treats education as investment: incur costs C now (direct costs plus forgone earnings) for a stream of extra earnings Δw_t later. Set the cost against the **present value** of that stream. The **internal rate of return** r^* is the discount rate at which the two exactly balance — the investment breaks even:

$$C = \sum_{t=1}^N \frac{\Delta w_t}{(1+r^*)^t}.$$

Education is worth undertaking when r^* exceeds the worker’s cost of funds — the same rule a firm uses to judge any investment project.

Method 1 (solving for the internal rate of return). When the extra earnings are a constant Δw for N years, the sum is an **annuity**, and

$$C = \Delta w \cdot a(r^*, N), \quad \text{where} \quad a(r, N) = \frac{1 - (1 + r)^{-N}}{r}$$

is the annuity factor. So $a(r^*, N) = C/\Delta w$; find the r^* that hits that factor, by trying values (the factor falls as r rises).

Example — is the degree worth it? — the number. With $C = €45000$, $\Delta w = €6000$ and $N = 40$, the break-even annuity factor is $a(r^*, 40) = 45000/6000 = 7.5$. Try a few rates:

$$a(10\%, 40) = 9.78, \quad a(13\%, 40) = 7.63, \quad a(14\%, 40) = 7.11.$$

The factor 7.5 sits just above 13%, so $r^* \approx 13\%$. Because 13% comfortably exceeds any realistic cost of funds (a student loan at, say, 5%), the degree is **worth it** — and the crude €240 000 figure overstated the case by ignoring both discounting and the timing of the €45 000 outlay. Note how the answer depends on the horizon: shorten the career to $N = 10$ years and $a(r^*, 10) = 7.5$ requires $r^* \approx 5.6\%$ — barely above the cost of funds. The young invest in schooling precisely because they have more years to amortise it.

3.2. Estimating the return on real data: the Mincer equation

Real-data returns are read off a **regression** — an object this chapter and several later ones lean on. Your prerequisite courses did not cover it, and this course never asks you to run one; but you must be able to **read** one, so here is the whole of what that takes.

Definition 7. A **regression** summarises how one variable moves with others by fitting the best straight line through a cloud of data points. In $\ln w = \beta_0 + \beta_1 X + \varepsilon$, the characteristic X (say years of schooling) is the input; the **slope** β_1 is the average change in the outcome for a one-unit rise in X — the **return** to that characteristic; the **intercept** β_0 is the fitted outcome when $X = 0$; and the **residual** ε collects everything about an individual the line does not capture. To **estimate** the equation is to choose β_0 and β_1 from data so that the residuals are collectively as small as possible. The estimation **mechanics** belong to the Data courses; here every coefficient is **given**, and all that is ever required of you is to read what a slope, an intercept, and a residual mean.

Definition 8. On real earnings data the return is estimated from the **Mincer equation**, a log-linear regression of earnings on years of schooling s and experience:

$$\ln w = \beta_0 + \rho s + \beta_1 \exp + \beta_2 \exp^2 + \varepsilon.$$

Note the letters: schooling carries its own coefficient ρ here, **not** the generic slope β_1 of the box above; ρ is the object of interest and reads as the average proportional earnings gain per extra year of schooling. The experience terms $\beta_1 \exp + \beta_2 \exp^2$ (with $\beta_2 < 0$) trace the familiar concave career profile — earnings rise with experience, then flatten; they are controls here, and their coefficients β_1, β_2 are not the return to schooling.

Remark. What the data say. Estimated Mincer returns cluster around $\rho \approx 0.07$ to 0.10 — roughly a 7–10% earnings gain per extra year of schooling — in the United States and much of Europe (Card’s 1999 survey of dozens of studies), with somewhat lower returns in a few high-schooling continental countries and higher returns in some developing economies where schooling is scarce. Because the equation is in logs, coefficients **add in log points** and only then convert to a percentage through $e^x - 1$ — not one-for-one. A ρ of 0.08 over a completed three-year degree adds $3 \times 0.08 = 0.24$ log points, an earnings gain of $e^{0.24} - 1 \approx 27\%$ (the linear shortcut “ $3 \times 8\% = 24\%$ ” understates it — the gap grows with the premium) — in the same range as the €6 000-on-a-mid-career-wage figure of the worked example.

Pitfall. Reading ρ as the **causal** return to schooling is the most common mistake here. People who get more schooling differ in unobserved ways (**ability**, family, motivation) that also raise earnings — **ability bias** — so the raw correlation overstates the causal return. Untangling the two is the original problem that drove labour economists toward instrumental variables (Chapter 9): using something that shifts schooling but not earnings directly — compulsory-schooling laws, distance to college, quarter of birth — to recover the causal effect.

3.3. Human capital versus signalling

The Mincer premium is a fact about **earnings**. It does not, by itself, tell you **why** the schooled are paid more — and two theories give opposite answers, with opposite policy conclusions.

Example — does the degree build the skill, or reveal it?. Two graduates command the earnings premium of the last section. In the first story the degree **raised their productivity**: they learned things employers value, and the premium is the market price of that added skill. In the second, the degree taught them nothing usable on the job — it merely **revealed** that they were already able, because only able people could get through it. The wage data look identical under both stories. The welfare conclusions could not differ more.

Definition 9. The **human-capital** view (Becker) holds that schooling **raises** productivity, so the earnings premium reflects real skill the education added. The **signalling** view (Spence) holds that schooling can raise earnings **even if it adds no productivity**, by serving as a credible **signal** of pre-existing ability. Signalling works only if the schooling is **cheaper to acquire for high-ability workers** (in effort, not just fees): that cost gap is what lets a degree **separate** the able from the rest, so that employers who cannot observe ability directly read the diploma as a proxy for it. Both views predict a positive schooling–earnings correlation; they disagree on what produces it.

Remark. Why the distinction is not academic. If schooling builds human capital, widening access to it raises productivity and is a social investment. If schooling is largely a signal, subsidising **more** of it for everyone is partly wasteful — it raises the credential bar without adding skill (a race for ever-higher credentials), and the **private** return to a degree exceeds its **social** return. The reality is some of each. The evidence usually cited for a signalling component is the

sheepskin effect: the earnings jump at the **completion** of a degree, over and above the same years of study without the diploma — hard to explain if only the learning mattered, natural if the **certificate** is what the market reads.

Remark. The macro test: does schooling cause growth? If schooling mainly builds productive human capital, countries that added years of schooling should have grown measurably faster. Bils and Klenow (2000) put this to the data and found the cross-country link is weak and, if anything, runs the **other** way: faster growth seems to induce more schooling more than schooling drives the growth. The finding does not overturn the micro Mincer premium — individuals who study more do earn more — but it warns that the **aggregate** productivity pay-off of mass schooling is smaller and harder to find than the **private** return, which is exactly the gap a signalling component would open: a credential can raise one worker's wage by sorting them ahead of others without adding to what the whole economy produces.

Pitfall. A positive Mincer coefficient is consistent with **both** readings; wage data **alone** cannot tell you which mechanism is at work. Telling them apart needs evidence of a different kind — a sheepskin effect, or a policy that changes credentials without changing learning — not a larger or more carefully estimated regression.

Exercises

3.1. A two-year professional degree costs €36 000 (fees plus forgone earnings) and yields €4 500 of extra earnings per year for 30 years. Use the annuity factors $a(11\%, 30) = 8.69$, $a(12\%, 30) = 8.06$, and $a(13\%, 30) = 7.50$. (a) Write the internal-rate-of-return equation and find the break-even annuity factor. (b) Bracket r^* with the factors given. (c) State the rule for deciding whether to invest, and say whether this degree passes it against a 6% cost of funds.

3.2. (a) Explain why the Mincer-equation coefficient on schooling overstates the **causal** return (ability bias) and how an instrumental variable such as quarter of birth addresses it. (b) The same degree premium fits both the human-capital and the signalling reading; state each reading and name one piece of evidence that would point toward signalling. (c) In one sentence, say what Bils and Klenow's cross-country finding adds to that debate.

Answers. (1) $36000 = 4500 \cdot a(r^*, 30)$, so the break-even factor is $a(r^*, 30) = 36000/4500 = 8$. Since $a(12\%, 30) = 8.06$ and $a(13\%, 30) = 7.50$, the factor 8 sits just below 12%, so $r^* \approx 12\%$. Invest when r^* exceeds the cost of funds; $12\% > 6\%$, so the degree passes comfortably.

(2) Ability bias: the schooled differ in unobserved ability that also raises earnings, so the OLS coefficient overstates the causal return; an instrument (quarter of birth, which shifts schooling through compulsory-schooling laws but has no direct path to earnings) recovers the causal effect. Human capital: schooling **raised** productivity, so the premium is added skill. Signalling: schooling **reveals** pre-existing ability without adding skill, working because a degree is cheaper for high-ability workers to complete; a **sheepskin effect** — an earnings jump at degree **completion**, beyond the same years of study without the diploma — points toward signalling. Bils and Klenow found schooling's aggregate growth pay-off weak, consistent with a signalling component alongside genuine human capital.

4. Wage structure: differentials, discrimination, decomposition

Two workers with the same schooling and experience are often paid differently. Before reaching for “discrimination,” an economist must rule out two competing explanations — and then learn to **measure** what is left.

4.1. Compensating differentials (Rosen)

Example — why the night shift pays more. Two warehouse workers, identical on paper. One works days at €18/hour; the other works nights, in the cold, at €22/hour. The €4 gap is not unequal treatment — it is the price the firm must pay to staff an unpleasant shift. Strip it out and the jobs pay the same.

Definition 10. Compensating differentials are wage differences that offset non-wage job characteristics. Unpleasant, dangerous, or insecure jobs must pay **more** to attract workers; desirable attributes are “paid for” through **lower** wages. Part of any observed wage gap may therefore reflect differences in job **attributes**, not in treatment of **workers**. Read this way, a wage is a **bundle of prices** — a base rate plus an implicit price for each job characteristic; recovering those implicit prices from wage data is the **hedonic** reading of wage differences.

4.2. Two kinds of discrimination

Economics distinguishes two mechanisms behind a group wage gap, and they call for different evidence and different remedies.

Definition 11. Taste-based discrimination (Becker) is a **preference** against a group: an employer, co-worker, or customer is willing to forgo money to avoid hiring or working with its members. It is prejudice in the literal sense — a distaste entered directly into the payoff. Under competition it is costly to the discriminator and can be eroded by firms that do not share the taste.

Definition 12. Statistical discrimination arises when employers cannot observe an individual’s productivity and use **group averages** as a proxy. Members of a group are then paid according to the group mean rather than their own productivity. The result is a persistent group wage gap **without any prejudice** — the employer is minimising an information problem, not indulging a taste — and it can be self-confirming, since the disadvantaged group may then under-invest in skills the employer cannot see.

4.3. The Oaxaca–Blinder decomposition

Example — splitting a wage gap. Men in a firm earn, on average, a log-wage of $\overline{\ln w_A} = 3.00$; women, $\overline{\ln w_B} = 2.75$. The raw gap is 0.25 log points (about 28%). How much of it is **explained** by men having more measured experience, and how much remains **unexplained**? The Oaxaca–Blinder decomposition answers exactly this — and below we run it to numbers.

Definition 13. The **Oaxaca–Blinder decomposition** splits a mean wage gap into an **explained** part (groups differ in characteristics) and an **unexplained** part (groups face different returns to the same characteristics). Estimate $\ln w = \beta_0 + \beta_1 X + \varepsilon$ separately in groups A and B , with mean characteristic $\bar{X}_A, \bar{X}_B$. Since $\overline{\ln w_g} = \hat{\beta}_{0,g} + \hat{\beta}_{1,g} \bar{X}_g$,

$$\underbrace{\overline{\ln w_A} - \overline{\ln w_B}}_{\text{raw gap}} = \underbrace{(\bar{X}_A - \bar{X}_B) \hat{\beta}_{1,B}}_{\text{explained (endowments)}} + \underbrace{(\hat{\beta}_{0,A} - \hat{\beta}_{0,B}) + \bar{X}_A(\hat{\beta}_{1,A} - \hat{\beta}_{1,B})}_{\text{unexplained (returns + constant)}}.$$

Example — the same gap, decomposed to numbers. Take experience as the one measured characteristic and estimate the wage equation in each group:

$$\text{men (A): } \ln w = 2.40 + 0.030 \cdot \text{exp}, \quad \overline{\text{exp}}_A = 20;$$

$$\text{women (B): } \ln w = 2.35 + 0.025 \cdot \text{exp}, \quad \overline{\text{exp}}_B = 16.$$

Check the means: $\overline{\ln w_A} = 2.40 + 0.030 \times 20 = 3.00$ and $\overline{\ln w_B} = 2.35 + 0.025 \times 16 = 2.75$, so the raw gap is 0.25. Now decompose, valuing the endowment gap at women’s returns $\hat{\beta}_{1,B} = 0.025$:

$$\text{explained} = (\overline{\text{exp}}_A - \overline{\text{exp}}_B) \hat{\beta}_{1,B} = (20 - 16) \times 0.025 = 0.10,$$

$$\text{unexplained} = (2.40 - 2.35) + 20 \times (0.030 - 0.025) = 0.05 + 0.10 = 0.15.$$

So 0.10 of the 0.25 gap (40%) is explained by men’s greater experience; 0.15 (60%) is **unexplained** — men are paid more even at equal experience, through both a higher intercept and a steeper return to each extra year. That 0.15 is what the debate is about.

Pitfall. The unexplained component is **not** a clean measure of discrimination. It also absorbs any productivity difference you failed to measure and any compensating differential you omitted, and it moves when you add or drop a control. A **statistical-discrimination** reading attributes it to group-based inference; a **taste-based** reading attributes it to prejudice; a **compensating-differentials** reading attributes part of it to unmeasured job attributes. The decomposition **frames** the debate; it does not settle it. (And the split is not unique: value the endowment gap at men’s returns instead and the two components shift — the “index-number problem.”)

Exercises

4.1. Two groups have wage equations $\ln w_A = 2.50 + 0.028 \cdot \text{exp}$ (mean experience 22) and $\ln w_B = 2.44 + 0.024 \cdot \text{exp}$ (mean experience 18). (a) Compute each group’s mean log wage and the raw gap. (b) Apply the Oaxaca–Blinder decomposition, valuing the endowment gap at group B ’s returns, to split the gap into explained and unexplained components. (c) Explain why the

unexplained component cannot be read directly as discrimination, distinguishing taste-based, statistical-discrimination, and compensating-differentials interpretations.

Answers. (1) Means: $\overline{\ln w_A} = 2.50 + 0.028(22) = 3.116$, $\overline{\ln w_B} = 2.44 + 0.024(18) = 2.872$, raw gap 0.244. Explained $= (22 - 18)(0.024) = 0.096$; unexplained $= (2.50 - 2.44) + 22(0.028 - 0.024) = 0.06 + 0.088 = 0.148$ (and $0.096 + 0.148 = 0.244$, checking). The 0.148 also holds unmeasured productivity, omitted compensating differentials, and moves with the controls, so it is not “discrimination” as such: a taste-based reading attributes it to prejudice, a statistical-discrimination reading to group-based inference, a compensating-differentials reading to unmeasured job attributes.

5. Unemployment: frictions, matching, and the natural rate

The supply-and-demand diagram has a glaring silence: at the market-clearing wage **everyone who wants a job has one**, so equilibrium unemployment is zero. Reality never is. The resolution is that hiring is not instantaneous — workers and jobs must **find** each other, and that takes time and resources.

5.1. Three kinds of unemployment

Definition 14.

- **Frictional**: short-term joblessness from the time it takes to match workers to jobs (search, mobility, labour-market entry). Present even in a booming economy.
- **Structural**: a **mismatch** between the skills or locations of available workers and of available vacancies. Persistent.
- **Cyclical**: a shortfall of jobs caused by deficient aggregate demand over the business cycle (the output gap from your macroeconomics course).

5.2. The Mortensen–Pissarides matching model

Example — hiring as a meeting process. A city has 100 000 unemployed workers and 80 000 open vacancies — yet nobody is hired **instantly**. Each day a certain number of workers and firms meet and pair up: more jobseekers and more vacancies both mean more matches, but search frictions mean the pool never clears at once. Modelling that meeting process — not a wage that equates two schedules — is the Mortensen–Pissarides contribution.

Definition 15. Search frictions are captured by a **matching function** giving the flow of new hires from the stocks of unemployment U and vacancies V :

$$M = m(U, V),$$

increasing in both and usually constant-returns-to-scale. Writing everything per unit of labour force (rates $u = U/L$, $v = V/L$), define **labour-market tightness** $\theta = v/u = V/U$. Constant returns let us write the **job-finding rate** — the rate at which an unemployed worker finds work — as

$$f(\theta) = \frac{M}{U} = m(1, \theta),$$

increasing in θ : a tighter market (more vacancies per jobseeker) fills the unemployed faster.

Now develop the model to its two payoffs: the equilibrium unemployment rate, and the Beveridge curve.

Proposition 2 (steady-state unemployment). Let jobs be destroyed at rate δ (separations per period). In steady state the flow **into** unemployment equals the flow **out**:

$$\underbrace{\delta(1-u)}_{\text{inflow: separations}} = \underbrace{f(\theta) \cdot u}_{\text{outflow: matches}}.$$

Solving for u ,

$$u = \frac{\delta}{\delta + f(\theta)}.$$

Unemployment is high when jobs break up fast (large δ) and when finding a new one is slow (small $f(\theta)$, i.e. a slack market).

Example — an unemployment rate from two flow rates. Suppose each month 2% of jobs end ($\delta = 0.02$) and an unemployed worker has a 30% chance of finding work ($f(\theta) = 0.30$). Then

$$u = \frac{0.02}{0.02 + 0.30} = \frac{0.02}{0.32} = 6.25\%.$$

Now let a recession halve the job-finding rate to $f = 0.15$: u jumps to $0.02/0.17 = 11.8\%$. Unemployment nearly doubles with no change in how fast jobs are destroyed — the outflow side alone produced the change. This is why economists watch the job-finding rate, not just layoffs.

5.3. Reading the Beveridge curve

The steady-state condition also pins down the shape economists actually observe. As $\theta = v/u$ rises, $f(\theta)$ rises, so $u = \delta/(\delta + f(\theta))$ **falls**: higher vacancies go with lower unemployment. That downward-sloping locus in (u, v) space is the **Beveridge curve**.

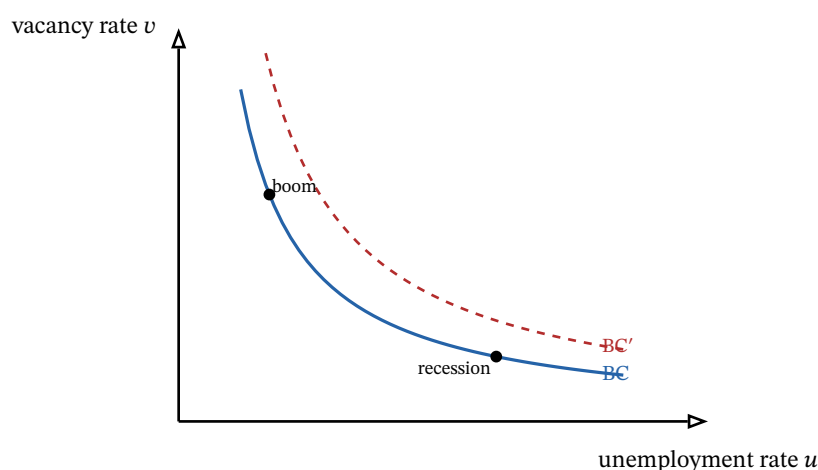


Figure 4: The Beveridge curve relates the vacancy rate to the unemployment rate. Movements **along** it trace the cycle: booms sit upper-left (many vacancies, low u), recessions lower-right. An **outward shift** to BC' — more unemployment at every vacancy rate — signals worse matching efficiency (a fall in f for given θ), i.e. a rise in **structural** unemployment. This is exactly what many economies observed after 2009 and again in 2021–22.

Pitfall. Read the **position**, not just the direction. **High vacancies together with high unemployment** is the diagnostic of a **matching/structural** problem — jobs exist but the unemployed cannot fill them (an outward-shifted curve). **Low vacancies with high unemployment** points to **deficient demand** (cyclical) — a slide down and to the right along the curve. Confusing the two leads to the wrong policy: retraining for a structural problem, demand stimulus for a cyclical one.

5.4. The NAIRU

Example — why pushing unemployment ever lower backfires. Suppose policy pushes unemployment down to 3% and tries to hold it there. At first vacancies are plentiful and wages climb. But with so few jobseekers per vacancy, firms bid ever harder for workers; wages and prices chase each other, and inflation does not merely rise — it keeps **accelerating**. There is a rate of unemployment below which this happens and above which inflation is stable. That threshold is what the next definition names.

Definition 16. The **NAIRU** (non-accelerating-inflation rate of unemployment) is the equilibrium unemployment rate consistent with **stable** inflation — the labour-market counterpart of the vertical long-run Phillips curve from your macroeconomics course. Frictional and structural unemployment **persist** at the NAIRU; cyclical unemployment is the deviation from it. Driving u below the NAIRU buys accelerating inflation, not a permanent jobs gain.

Remark. Orders of magnitude. The NAIRU is estimated, not observed, and it drifts. US estimates (e.g. the Congressional Budget Office) put the long-run natural rate around 4.5% in the 2010s–2020s, down from 6%-plus in the 1980s; euro-area estimates are higher and more dispersed, commonly 7–9%, reflecting the matching and institutional differences of the next chapter.

Exercises

5.1. In a labour market jobs end at rate $\delta = 0.025$ per month and the job-finding rate is $f(\theta) = 0.20$. (a) Compute steady-state unemployment. (b) A recession halves the job-finding rate; recompute. (c) State the Beveridge curve and read three scenarios — high vacancies with high unemployment, low vacancies with high unemployment, and an outward shift of the whole curve — classifying each as frictional, cyclical, or structural.

Answers. (1) $u = 0.025/(0.025 + 0.20) = 0.025/0.225 = 11.1\%$; with $f = 0.10$, $u = 0.025/0.125 = 20.0\%$ — unemployment nearly doubles though δ is unchanged, the outflow side alone producing the change. Beveridge curve (Figure 4): high v + high $u \rightarrow$ structural (mismatch); low v + high $u \rightarrow$ cyclical (deficient demand); an outward shift $\rightarrow$ worse matching, i.e. higher structural unemployment.

6. Institutions and the minimum-wage debate

Labour markets are never left bare: minimum wages, unions, and unemployment insurance sit on top of them in every advanced economy. The competitive model of Chapters 1 and 2 makes a sharp — and sharply **negative** — prediction about each. We put that case first, read straight off the supply-and-demand diagram you already have, and only then ask why every one of these institutions is adopted regardless.

6.1. The competitive case against wage floors and insurance

First, a term the insurance argument leans on.

Definition 17. A worker's **reservation wage** is the lowest wage at which they will accept a job rather than keep searching. Anything that raises the value of remaining unemployed — more generous benefits, better prospects — raises the reservation wage, and a higher reservation wage means longer search. (This is the search counterpart of the **participation** reservation wage of Chapter 1: there, the wage below which a worker supplies no hours; here, the wage below which a searcher rejects an offer and keeps looking.)

Now run each institution through the model, shifting the very curves the course has built (Figure 5).

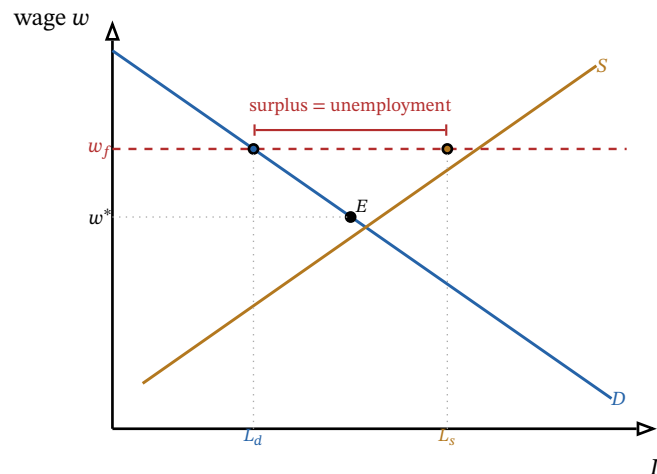


Figure 5: The competitive case against a wage floor. A minimum wage or a union wage w_f set above the market-clearing w^* moves the firm **up its demand curve** to L_d while drawing **more workers up the supply curve** to L_s . The gap $L_s - L_d$ is a labour surplus — involuntary unemployment. Unemployment insurance works on the same diagram from the other side: by raising the reservation wage it shifts **supply** leftward (fewer accept work at each wage), lifting the equilibrium wage and trimming employment.

Proposition 3. In the competitive model each institution **worsens** employment:

- **Minimum wage:** a binding minimum lifts wages but cuts employment; the size of the cut depends on labour-demand elasticity.

- **Unions:** collective bargaining raises wages above the competitive level for members (the **union wage premium**), which can lower union-sector employment and push workers into the non-union sector.
- **Unemployment insurance:** replacement income during search raises the **reservation wage** and lengthens search, raising frictional unemployment — while insuring consumption and potentially improving match quality.

6.2. Why these institutions exist all the same

If the diagram is so damning, why does every rich democracy keep wage floors, recognise unions, and insure the unemployed? Because the competitive diagram is not the whole model, and because employment is not the only thing at stake.

Collective bargaining as countervailing power. The competitive case assumes firms are wage-takers. Where an employer has **monopsony** power — the case developed in the next section — a single worker bargains from weakness, and a union (or a wage floor) restores a balance the lone worker cannot. Read this way, collective bargaining is not a distortion added to a competitive market but a **correction** to an uncompetitive one: it can raise the wage **and** employment over exactly the range in which a minimum wage can, for the same monopsony reason. Unions also give workers a **voice** — a channel to improve conditions and reduce costly turnover — which is why unionised firms are not uniformly less productive.

The political economy. Insurance against job loss is a good private markets underprovide: a worker cannot buy a policy against a recession, because the risk hits everyone at once and those most exposed are the least able to pay. The state steps in where the private market fails. And workers are not only agents in a model but a **political coalition**: they vote, and in a democracy the median voter is a wage-earner who values income security and a wage floor. Labour-market institutions are, in the end, the outcome of that political economy as much as of any efficiency calculation — which is why they persist even where the textbook diagram counts them a cost.

Remark. **Comparative data complicate the simple story.** The countries with the highest wage floors and broadest bargaining coverage do not uniformly show the highest unemployment:

Country	Min. wage (2024)	Bargaining coverage	Union density	Unemp. 2023
United States	\$7.25/hr	12%	10%	3.6%
France	€11.65/hr	98%	8%	7.3%
Germany	€12.41/hr	54%	16%	3.0%
Denmark	none*	82%	67%	5.1%

*Denmark sets wage floors through sectoral bargaining, not statute. Denmark pairs generous insurance and high coverage with light hiring and firing rules and moderate unemployment — the **flexicurity** model. Institutions interact; no single one reads off equilibrium unemployment by itself. (Figures are indicative, drawn from OECD and Eurostat compilations.)

6.3. The Card–Krueger reframing and monopsony

Example — New Jersey, 1992. On 1 April 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05; neighbouring Pennsylvania did not. Card and Krueger surveyed over 400 fast-food restaurants on both sides of the border, before and after. The competitive model predicted New Jersey employment should fall relative to Pennsylvania. It did not: measured full-time-equivalent employment per restaurant **rose** slightly in New Jersey while **falling** in Pennsylvania — a difference of roughly +2.75 workers per restaurant in New Jersey’s favour, statistically indistinguishable from zero and certainly not the sharp fall predicted. This single well-designed comparison changed economists’ view of the minimum wage more than years of theoretical argument had.

Why might a minimum wage **fail** to cut jobs — or even raise them? Because the competitive model assumes firms are wage-**takers**. If instead an employer has **monopsony** power (it is large relative to its local labour market), it restrains hiring to hold wages down, and a minimum wage can push **both** wage and employment up over a range.

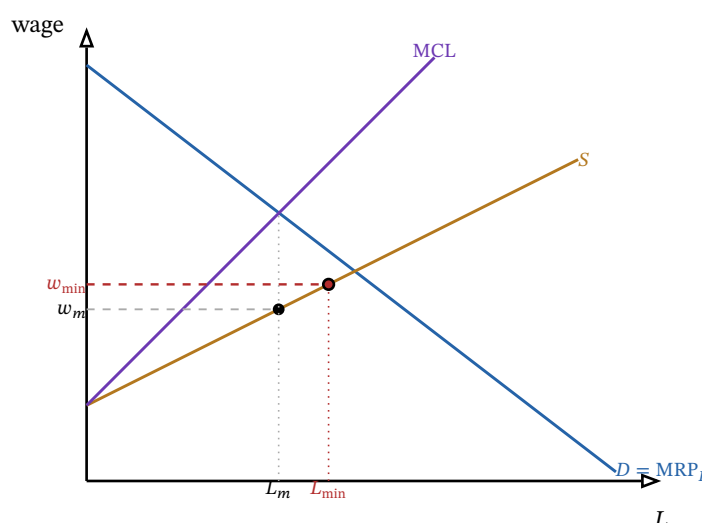


Figure 6: Minimum wage under **monopsony**. Unconstrained, the firm equates the marginal cost of labour (MCL, above supply S because hiring one more worker raises the wage paid to *all*) with demand, hiring only L_m at the low wage w_m . A minimum wage $w_{\min}$ set between w_m and the competitive wage flattens the firm’s marginal cost at $w_{\min}$: it becomes a wage-taker and hires every worker willing to work at the floor, out to where the flat $w_{\min}$ line meets **supply**, at $L_{\min} > L_m$ (the marginal product there still exceeds $w_{\min}$, so the firm wants them all). **Both wage and employment rise** — the result the competitive model cannot produce, and the theory Card–Krueger revived.

Remark. What the natural-experiment evidence changed, in three registers:

- **Empirical:** modest minimum-wage rises have small or negligible employment effects, contradicting the naïve competitive prediction.
- **Theoretical:** it revived **monopsony** (employer wage-setting power) as a serious description of low-wage labour markets.
- **Methodological:** it helped install the **natural-experiment** standard now expected of any causal claim in the field — the subject of Chapter 9.

Pitfall. The lesson is **not** “minimum wages never cost jobs.” It is that the employment effect is an **empirical magnitude**, not a theorem — and that it depends on how far the floor is pushed and on how competitive the local market is. A minimum wage set far above productivity will still destroy jobs; the monopsony range in Figure 6 is bounded (push $w_{\min}$ above the competitive wage and employment falls again).

Exercises

6.1. Using the competitive model, explain how a binding minimum wage, a union wage premium, and unemployment insurance each affect equilibrium unemployment (define the reservation wage in your answer). Then explain, with the monopsony diagram, how a minimum wage can raise **both** wage and employment, and state precisely what the Card–Krueger natural-experiment evidence changed — empirically, theoretically, and methodologically.

Answers. (1) Reservation wage: lowest wage a searcher will accept. Minimum wage above market-clearing → surplus/unemployment; union premium → lower union-sector employment; insurance → higher reservation wage, longer search, more frictional unemployment. Under monopsony (Figure 6) a minimum wage between w_m and the competitive wage raises both wage and employment. Card–Krueger changed the empirical verdict (small/zero effect), revived monopsony theory, and set the natural-experiment methodological standard.

7. Inequality measurement

To study the distribution we first need to **measure** it. Two tools — one visual, one a single number — do almost all the work, and both are built directly from a ranked list of incomes.

7.1. The Lorenz curve

Example — five households. Five households earn €10k, €20k, €30k, €40k, €100k — total €200k. Rank them poorest-first and accumulate. The poorest 20% (one household) hold €10k = 5% of income; the poorest 40% hold €30k = 15%; the poorest 60%, €60k = 30%; the poorest 80%, €100k = 50%; and all 100% hold 100%. Plot those cumulative pairs (Figure 7) and you have drawn this society's Lorenz curve — and the sag below the diagonal **is** its inequality.

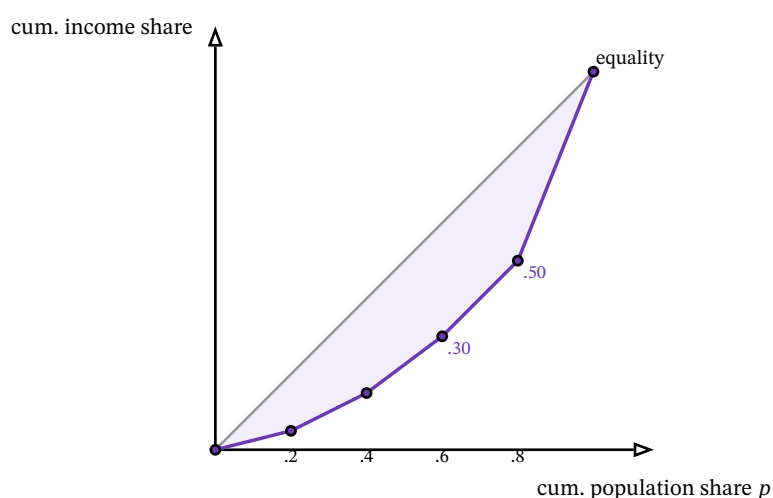


Figure 7: The Lorenz curve for the five households of the worked example, through the computed points $(0.2, 0.05)$, $(0.4, 0.15)$, $(0.6, 0.30)$, $(0.8, 0.50)$ and $(1, 1)$. The shaded gap between the diagonal and the curve is the raw material of the Gini coefficient computed opposite.

Definition 18. Rank the population poorest to richest. The **Lorenz curve** $L(p)$ plots the cumulative share of total income held by the poorest fraction p . It runs from $(0, 0)$ to $(1, 1)$, lies on or below the 45° line of perfect equality, and is convex. The farther it sags below the diagonal, the greater the inequality.

7.2. The Gini coefficient

Definition 19. The **Gini coefficient** measures inequality as the area between the equality line and the Lorenz curve. Write A for the area between the diagonal and the curve and B for the area beneath the curve. Then

$$G = \frac{A}{A + B}.$$

The whole triangle under the diagonal has area $A + B = 1/2$, so $G = A/(1/2) = 2A$: the Gini is **twice** the gap between the curve and the diagonal. It ranges from 0 (perfect equality — the curve lies on the diagonal, so $A = 0$) to 1 (perfect inequality — one household holds everything). The method below turns this area into a formula you can compute from grouped data.

Method 2 (the Gini from grouped data). With cumulative population shares $p_0 = 0, \dots, p_n = 1$ and Lorenz ordinates $L_0 = 0, \dots, L_n = 1$, approximate the area under the curve by trapezoids:

$$G \approx 1 - \sum_{i=1}^n (p_i - p_{i-1})(L_i + L_{i-1}).$$

Example — the Gini for the five households. Use the five-household ordinates from the worked example above. Each population step is $p_i - p_{i-1} = 0.2$, so

$$\begin{aligned} G &\approx 1 - 0.2 \times [(0.05 + 0) + (0.15 + 0.05) + (0.30 + 0.15) + (0.50 + 0.30) + (1.00 + 0.50)] \\ &= 1 - 0.2 \times [0.05 + 0.20 + 0.45 + 0.80 + 1.50] = 1 - 0.2 \times 3.00 = 1 - 0.60 = 0.40. \end{aligned}$$

This society's Gini is 0.40 — squarely in the range of a fairly unequal advanced economy. The whole chapter's machinery now delivers a single number you can compare across countries and years.

Example — the same computation on real data — the United States. The method transfers unchanged to an actual country. The World Bank reports US household income shares by quintile for 2024 as 5.1%, 10.0%, 15.1%, 22.6% and 47.1% (poorest fifth to richest). Accumulate them into Lorenz ordinates

$$L = 0.051, 0.151, 0.302, 0.528, 1.000$$

and feed those to the very same trapezoidal formula (step 0.2 again):

$$\begin{aligned} G &\approx 1 - 0.2 \times [(0.051 + 0) + (0.151 + 0.051) + (0.302 + 0.151) + (0.528 + 0.302) + (1.000 + 0.528)] \\ &= 1 - 0.2 \times 3.064 = 1 - 0.613 = 0.39. \end{aligned}$$

The World Bank's own published Gini for the United States is 0.42. Our 0.39 falls a little short **for a reason worth seeing**: five coarse groups force the Lorenz curve to be a straight line inside each quintile, which erases the inequality **within** the top fifth — exactly the tail the Gini is least sensitive to (the Pitfall below). Finer data — deciles, or percentiles at the very top — would close most of the 0.03 gap. Real distributional data thus give both the number and a lesson in how grouping biases it.

Remark. The trajectory since 1980. On real disposable-income (post-tax, post-transfer) data, the Gini has **risen** in many advanced economies since around 1980 — most sharply in the United States (from roughly 0.35 around 1980 to about 0.39–0.41 today) and the United Kingdom (to about 0.35), more mildly in France (around 0.29–0.32), and it remains lowest in the Nordic economies (Denmark, Sweden around 0.25–0.28). The dispersion across countries facing the **same** global forces (trade, technology) is itself evidence that **institutions and redistribution** — not technology alone — shape the outcome.

Pitfall. The Gini compresses an entire distribution into one number, so very different distributions can share a Gini. It is relatively insensitive to the **tails** — two societies with the same Gini can differ enormously in their **top 1% share**. For top-end concentration (the Piketty agenda of Chapter 8) report top shares alongside the Gini; no single statistic is sufficient.

7.3. Percentile ratios and top shares

The Gini reads the **whole** curve into one number. Two simpler statistics read **named ranks** instead, and they answer questions the Gini blurs.

Example — the gap between two ranks. Line everyone up by income. The person standing at the 90th percentile earns €6 000 a month; the person at the 10th earns €1 200. The **P90/P10 ratio** is $6000/1200 = 5$: the top of the middle earns five times the bottom. One number, from two points on the distribution — and it says something the Gini cannot, namely how far apart these two particular ranks stand.

Definition 20. A **percentile ratio** compares incomes at two ranks of the distribution. The **P90/P10 ratio** divides the 90th-percentile income by the 10th; **P50/P10** compares the median to the bottom decile (lower-half inequality); **P90/P50** the top to the median (upper-half inequality). A percentile ratio is **local** — it uses only the two ranks named and ignores everything between and beyond them — which is exactly why it is reported **alongside** the Gini, not instead of it: it pins down **where** in the distribution the stretching happens.

Example — splitting inequality into halves. With monthly incomes $P10 = €1200$, $P50 = €2400$, $P90 = €6000$:

$$P90/P10 = \frac{6000}{1200} = 5, \quad P50/P10 = \frac{2400}{1200} = 2, \quad P90/P50 = \frac{6000}{2400} = 2.5.$$

Because $P90/P50 = 2.5 > 2 = P50/P10$, this distribution is stretched more at the **top** than at the bottom — the gap above the median exceeds the gap below it. A single Gini would have hidden which half was doing the work.

Definition 21. A **top income share** is the fraction of total income received by a top group — the top 10%, top 1%, or top 0.1%. Because it targets precisely the **tail** the Gini is least sensitive to, it is the natural second number to report beside the Gini.

Remark. **Why fiscal data revolutionised top shares.** Top shares were long estimated from household **surveys**, which systematically **under-catch the very top**: the richest households are few and hard to sample, and they tend to under-report capital income. Piketty, Saez and their collaborators rebuilt the series from **tax records** instead — administrative data that cover the whole population, top included — now assembled for many countries in the **World Inequality Database**. The switch revealed top shares markedly higher, and rising faster since 1980, than

surveys had shown: the United States top-1% pre-tax income share roughly **doubled**, from about 10% around 1980 to above 20% by the 2010s.

Exercises

7.1. Five households earn €15k, €25k, €35k, €45k, and €80k. (a) Construct the Lorenz ordinates. (b) Compute the Gini coefficient using the trapezoidal formula. (c) Interpret the result and explain why two distributions can share a Gini yet differ sharply in top-1% share. (d) For monthly incomes $P_{10} = €1500$, $P_{50} = €3000$, $P_{90} = €9000$, compute P_{90}/P_{10} , P_{50}/P_{10} , and P_{90}/P_{50} , and say which half of the distribution is more stretched. (e) Pick one country and, using this chapter's trajectory figures, describe how its Gini has moved since 1980; then give the leading interpretation of **why**, and explain what the divergence between countries facing the **same** global forces tells you about the role of institutions.

Answers. **(1)** Total €200k, so ordinates (p, L) : $(0.2, 0.075)$, $(0.4, 0.20)$, $(0.6, 0.375)$, $(0.8, 0.60)$, $(1, 1)$. $G \approx 1 - 0.2(0.075 + 0.275 + 0.575 + 0.975 + 1.60) = 1 - 0.2 \times 3.50 = 1 - 0.70 = 0.30$ — less unequal than the €10k–€100k spread of the worked example (0.40), because the top household is nearer the rest. Two distributions share a G yet differ in the top-1% share because G is tail-insensitive. **(d)** $P_{90}/P_{10} = 9000/1500 = 6.0$, $P_{50}/P_{10} = 3000/1500 = 2.0$, $P_{90}/P_{50} = 9000/3000 = 3.0$; since $3.0 > 2.0$, the upper half is more stretched than the lower. **(e)** The United States, say: its disposable-income Gini rose from roughly 0.35 around 1980 to about 0.39–0.41 today — a clear widening. The leading reading is that skill-biased technology and globalisation pushed up top incomes across advanced economies, but redistribution and labour-market institutions (unions, minimum wages, taxes and transfers) governed how much of that pressure reached **disposable** incomes. France (≈ 0.29 – 0.32) and the Nordics (≈ 0.25 – 0.28) faced the same global forces yet saw far smaller rises — the divergence is itself evidence that institutions, not technology alone, shape the outcome.

8. Wealth inequality and its dynamics

Income is a flow; **wealth** is the stock that accumulates from it, and wealth is far more concentrated than income (top-10% wealth shares of 60–70% are common, against top-10% income shares nearer 30–40%). Piketty's contribution was to give the long-run dynamics of that concentration a disciplined framework.

8.1. Two distinct statements: the aggregate ratio and the concentration mechanism

It is easy to blur two of Piketty's claims. Keep them apart.

Proposition 4 (the aggregate capital/income ratio). If a country saves a fraction s of income and income grows at rate g , then in the long run the **aggregate** capital/income ratio converges to

$$\beta = \frac{s}{g}.$$

This is Piketty's "second fundamental law." It is a statement about the **size** of the capital stock relative to national income — **not** about who owns it. With $s = 10\%$ and $g = 2\%$, $\beta = 5$ (capital worth five years of national income); if growth slows to $g = 1.5\%$, $\beta = 6.7$. **Slower growth raises the weight of past wealth in the economy.**

The second claim is best met through a case before its statement.

Example — two families over a generation. The Capital family owns €1m returning $r = 5\%$ a year and reinvests it; after 30 years their fortune is $1\text{m} \times 1.05^{30} \approx \text{€}4.32\text{m}$. The Labour family's wages grow with the economy at $g = 1.5\%$; over the same 30 years their earnings rise only by a factor $1.015^{30} \approx 1.56$. The ratio of the two grows by $1.05^{30}/1.015^{30} = (1.05/1.015)^{30} \approx 2.77$: inherited wealth nearly triples relative to earned income, for no reason other than arithmetic — $r > g$. Historically Piketty puts r around 4–5% and g around 1–2% over the long run, with the gap widening as growth slows.

Proposition 5 (the $r > g$ concentration mechanism). Let r be the (after-tax) rate of return on wealth and g the growth rate of average income. A wealth-holder who reinvests sees their fortune grow at about r , while incomes grow at g . When

$$r > g,$$

large existing fortunes pull **away** from average incomes, so wealth **concentrates at the top** over time — a statement about the **distribution** of wealth, driven by the gap $r - g$, not by the aggregate ratio β .

Pitfall. Two traps here. First, do **not** fuse the two propositions: $\beta = s/g$ is about the **aggregate** stock and says nothing by itself about **top** shares; concentration at the top is what $r > g$ drives.

Second, $r > g$ is a **tendency**, not a law of motion that runs untouched. It is offset by the dispersion of returns, by wars and crises that destroy capital, by taxation, by consumption out of wealth, and by how estates are split across heirs. Piketty's claim is that **absent** such offsets the drift is toward concentration — not that concentration rises monotonically regardless of policy.

8.2. Intergenerational mobility and the Great Gatsby Curve

Definition 22. **Intergenerational mobility** measures how strongly a child's economic position depends on the parents'. It is summarised by the **intergenerational elasticity of earnings** (IGE) — the regression coefficient of child on parent log-earnings. A **higher** elasticity means a child's outcome is more tightly bound to the parents', i.e. **lower** mobility. Estimated IGEs run from about 0.15 in the Nordic countries (high mobility) to roughly 0.4–0.5 in the United States and United Kingdom (low mobility).

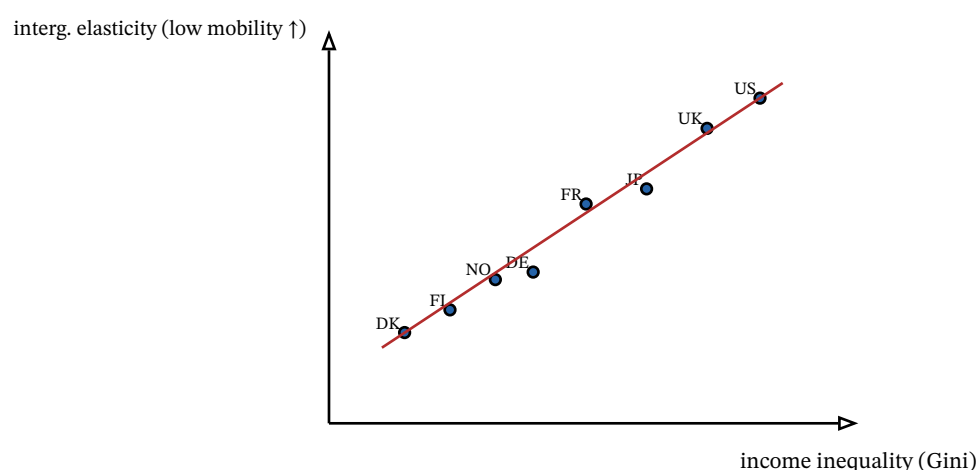


Figure 8: The **Great Gatsby Curve**: across countries, higher income inequality goes with a higher intergenerational elasticity — i.e. **lower** mobility. More unequal societies (US, UK) tend to transmit advantage across generations more strongly than more equal ones (Nordics). Inequality in one generation and immobility across generations reinforce each other. (Country positions are schematic but ordered as the data order them.)

Pitfall. The Great Gatsby Curve is a **cross-country correlation**, not a proven causal chain. It is consistent with inequality **causing** immobility (the rich buy decisive advantages for their children) but also with common third causes. Stating it as “inequality reduces mobility” without an identification strategy is exactly the inference error the next chapter exists to discipline.

8.3. Is inequality really rising? Two objections

The Ginis and top shares of the last two chapters point one way: within advanced economies, inequality has **risen** since 1980. Two objections are routinely raised against reading too much into that. Both deserve an answer — and answering them sharpens the claim rather than dissolving it.

Objection 1: mobility. A snapshot Gini, the argument runs, overstates **lifetime** inequality, because people **move** between ranks — today’s poor student is tomorrow’s earner. If ranks reshuffle each generation, a high Gini need not mean a rigid society. The answer is the Great Gatsby Curve just drawn: empirically, **mobility is lowest exactly where inequality is highest**. The reshuffle that would offset a high Gini is weakest in the most unequal places, so inequality and immobility **compound** rather than cancel. Mobility does not rescue the optimistic reading; it deepens the pessimistic one.

Objection 2: the elephant curve. Widen the lens from one country to the whole world and **global** inequality between individuals has, on the standard measure, **fallen**: hundreds of millions in Asia climbed out of poverty. The evidence is the **elephant curve** (Figure 9) — real-income growth from 1988 to 2008 plotted against a person’s rank in the **global** distribution, tracing the silhouette of an elephant. The answer is to read the **whole** animal. Its raised **body** — the global middle class, overwhelmingly in emerging Asia — did enjoy the fastest growth, and that is the genuine good news. But its lowered **trunk** shows the lower-middle class of **rich** countries nearly stagnant, and its raised **tip** shows the **global top 1%** capturing an outsized share: the top 1% is the main winner of globalisation. Between-country convergence and **within**-country divergence run at once — which is exactly why the within-country inequality this course measures did not go away.

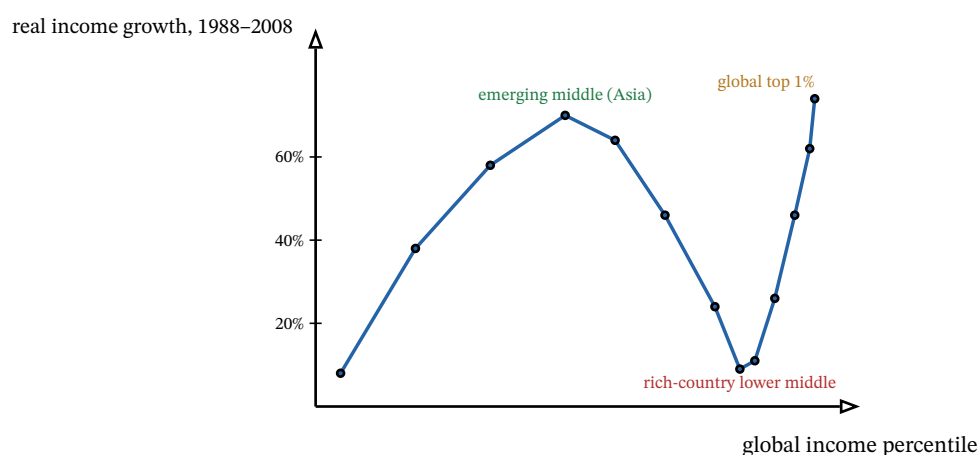


Figure 9: The **elephant curve**: real-income growth over 1988–2008 by global income percentile. The raised **body** around the median is the emerging-Asia middle class — the fastest gains; the lowered **trunk** near the 80–85th percentile is the lower-middle class of rich countries, nearly stagnant; the raised **tip** is the global top 1%, the outsized winner. Global inequality fell even as rich-country inequality rose. (Schematic, after Lakner and Milanovic.)

8.4. Why inequality matters: efficiency, not only fairness

Whether inequality is **unfair** is a moral question, and economics does not settle those. But there is a second, entirely positive question an economist can address: does inequality make the economy **less productive**? Past some level, the evidence says it can, through five channels.

Framework 1 (the efficiency costs of inequality).

1. **Innovation.** Talented children of poor families rarely become inventors — the “lost Einsteins”: exposure and opportunity, not raw ability, gate who patents. Steep inequality wastes inventive talent and lowers the innovation rate.

2. **Civic trust.** More unequal societies report lower interpersonal and institutional trust. Trust is the lubricant of exchange; where it erodes, contracting, enforcement, and cooperation all cost more.
3. **Infrastructure.** When the affluent secede into private substitutes — private schools, security, transport — political support for shared **public goods** weakens, and infrastructure with high social returns is under-provided.
4. **Education.** Credit constraints and unequal schooling stop able children of poor families from investing in their own human capital, so the economy's stock of skill is **misallocated** — a direct productivity loss.
5. **Safety.** Higher inequality is associated with more crime, diverting real resources into protection and enforcement that produce nothing.

None of this says **all** inequality is inefficient — some wage spread is the very incentive that draws effort and investment. The claim is narrower and firmer: past some level, inequality stops paying for itself in incentives and starts costing output through these five channels. That is why “inequality matters” is not only a value judgement.

Exercises

8.1. (a) State and distinguish Piketty's two propositions: the aggregate capital/income ratio $\beta = s/g$ and the top-concentration mechanism $r > g$, explaining why they are not the same claim; compute β for a country saving 12% of income that grows at 2%. (b) With $r = 6\%$ and $g = 2\%$, compute how much a reinvested fortune grows relative to average earnings over 25 years. (c) State the two objections to the claim that inequality is rising — mobility and the elephant curve — and give the answer to each. (d) Name three of the five efficiency costs of inequality and state each mechanism in one line.

Answers. (1) $\beta = s/g$ is the **aggregate** stock relative to income (size, not ownership): here $\beta = 12/2 = 6$. $r > g$ drives **top** concentration (distribution). Over 25 years a reinvested fortune grows relative to earnings by $(1.06/1.02)^{25} \approx 2.6 \times$. Mobility objection: it **fails**, because mobility is lowest where inequality is highest (Great Gatsby, Figure 8), so the two compound. Elephant-curve objection (Figure 9): read the whole curve — the global middle rose, but the global top 1% is the main winner and within-country inequality still widened. Efficiency costs (any three): innovation (wasted inventive talent), civic trust (higher transaction costs), infrastructure (under-provided public goods), education (misallocated human capital), safety (resources diverted to crime and protection).

9. Causal inference for labour-market policy

Every chapter so far has bumped into the same wall: a policy and an outcome move together, but does the policy **cause** the outcome? This chapter makes the methods explicit. They are the reason labour economics looks the way it does today, and they are the most transferable skill in the course.

9.1. The counterfactual problem

Example — the fundamental problem. A region raises its training subsidy and unemployment falls 2 points. Did the subsidy do it? To know, you would compare the region's actual outcome with what **the same region** would have done **without** the subsidy — but that world is never observed. The missing counterfactual is the whole difficulty; a raw before/after number silently assumes nothing else changed (it always does).

Definition 23. The **causal effect** of a treatment on a unit is the difference between its outcome **with** and **without** treatment — but only one is ever observed. This **fundamental problem of causal inference** means a naïve treated-vs-untreated or before-vs-after comparison **confounds** the policy with every other difference between the comparison groups or periods. All credible methods are devices for **constructing a believable counterfactual**.

One method **builds** the counterfactual outright; the rest **hunt** for one in data the world happened to produce. We take them in that order — from the manufactured counterfactual to the ever-cleverer ways of finding a ready-made one.

9.2. Randomised controlled trials: manufacturing a counterfactual

Example — a training programme by lottery. A city can fund a training place for only half of its 2 000 applicants. Rather than pick, it draws lots. Winners and losers are then, in expectation, identical in ability, motivation, and circumstance — the lottery guarantees it — so the losers' later employment **is** the winners' missing counterfactual. Any gap between the two groups is the programme's effect, with nothing left to confound it.

Definition 24. A **randomised controlled trial** (RCT) assigns the treatment **at random** — by lottery — so that treated and control groups are, in expectation, identical in every respect but the treatment. Randomisation **builds** the missing counterfactual rather than searching for one. It is therefore the **benchmark** against which every method below is judged: each is credible exactly to the degree it approximates random assignment.

Pitfall. The benchmark is not the whole answer, on two counts.

- **Feasibility.** Many labour-market questions cannot be randomised at all — you cannot assign a country its minimum wage, or a generation its level of inequality.

- **Validity.** A clean trial settles **internal** validity — inside the experiment the effect is real — by construction. It does **not** settle **external** validity — whether the same effect holds at national scale or in another country. Run on volunteers in one city, a result may not generalise; and people who know they are studied may behave differently.

This is why the field pairs trials with the observational methods below, each covering the other's weakness.

9.3. Natural experiments: finding a counterfactual

When randomisation is impossible, the next best thing is a counterfactual the world produced by accident.

Definition 25. A **natural experiment** exploits variation in a policy generated by an external event — a new law, an administrative border, a lottery, a date cutoff — that is **plausibly unrelated** to the outcome's other determinants, so that being treated is “as good as random.” Card and Krueger's state border is the canonical case: New Jersey raised its minimum wage and neighbouring Pennsylvania did not, so a fast-food job on the Pennsylvania side is a credible counterfactual for one on the New Jersey side. We compute that very comparison, formally, as difference-in-differences below.

9.4. Instrumental variables and regression discontinuity: building a clever counterfactual

Sometimes no ready-made control group exists, but a **sliver** of the treatment is assigned as-good-as-randomly by something outside the outcome. Two designs turn that sliver into a counterfactual.

Example — quarter of birth — a return to schooling that dodges ability bias. To measure the return to schooling free of ability bias, Angrist and Krueger needed something that shifts how long people stay in school but is otherwise unrelated to their ability or family. They found it in an unlikely place: **quarter of birth**. Because compulsory-schooling laws tie the legal leaving age to a birthday, men born early in the calendar year reach that age with slightly **less** schooling than those born later — for a reason, the month of birth, seemingly unrelated to earnings. Suppose the first-quarter-born average 12.7 years of schooling and log-earnings of 4.60, while the later-born average 12.8 years and 4.61. Take the gap in earnings between the two groups and divide by the gap in their schooling:

$$\text{effect} = \frac{4.60 - 4.61}{12.7 - 12.8} = \frac{-0.01}{-0.10} = 0.10.$$

A 10% causal return per extra year of schooling. Because it rests **only** on the schooling variation that birth timing drives — not on the choice of the more able to study longer — it sidesteps the **ability bias** that inflates the raw Mincer coefficient of Chapter 3.

Definition 26. The quarter-of-birth trick has a general name. An **instrumental variable** Z affects the treatment D but influences the outcome Y **only through** D . The two conditions are **relevance** (Z shifts D) and **exclusion** (Z has no other path to Y). When Z is a simple yes/no split of the sample, it recovers the causal effect as a **ratio of two differences in group means** — the gap in the outcome between the two Z -groups, divided by the gap in the treatment they receive:

$$\text{effect} = \frac{\bar{Y}_{Z=1} - \bar{Y}_{Z=0}}{\bar{D}_{Z=1} - \bar{D}_{Z=0}}.$$

The numerator is the instrument's whole effect on the outcome; dividing by its effect on the treatment rescales it into an effect **per unit of treatment**. Both differences are ordinary averages — the estimator asks for nothing beyond group means you can read straight off the data. In the example, Z is quarter of birth, D years of schooling, and Y log-earnings.

Pitfall. Everything rests on the **exclusion restriction** — and it is an **assumption** no data can confirm. The quarter-of-birth estimate is the causal return **only if** birth timing reaches earnings through **no channel but schooling**. Name the ways it might not: children born in different seasons start school at different **relative ages**, which can affect attainment and confidence on their own; and season of birth is mildly correlated with maternal health and family background, which affect earnings directly. If any such second door exists, the numerator picks it up too and the ratio no longer identifies the return to schooling. A **weak** instrument — here the denominator is a tenth of a year — makes this worse, because it magnifies any leak in the exclusion restriction. An instrument is only ever as good as the argument that it has no second path to the outcome.

Example — Maimonides' rule — the effect of class size. An old rule invoked in Israeli schools caps a class at 40: enrol 40 pupils and they sit in one class of 40; enrol 41 and the grade splits into two classes of about 20. Average class size therefore **plunges** each time enrolment crosses a multiple of 40 — a sharp, mechanical discontinuity with nothing to do with how good or rich a school is. Angrist and Lavy compare achievement in schools **just above** a cutoff (small classes, by the rule) with those **just below** (large classes): two schools enrolling 39 and 41 are alike in everything a parent would choose on, yet land in classes of 39 and 21 — as good as a randomly assigned class size. The jump in achievement at the cutoff is the effect of class size.

Definition 27. A **regression-discontinuity** (RD) design generalises Maimonides' rule: it exploits a **threshold** in a rule where units just below and just above a cutoff are near-identical, yet only those past it are treated, so the **jump** in the outcome at the cutoff — not the overall slope — is the causal effect (Figure 10).

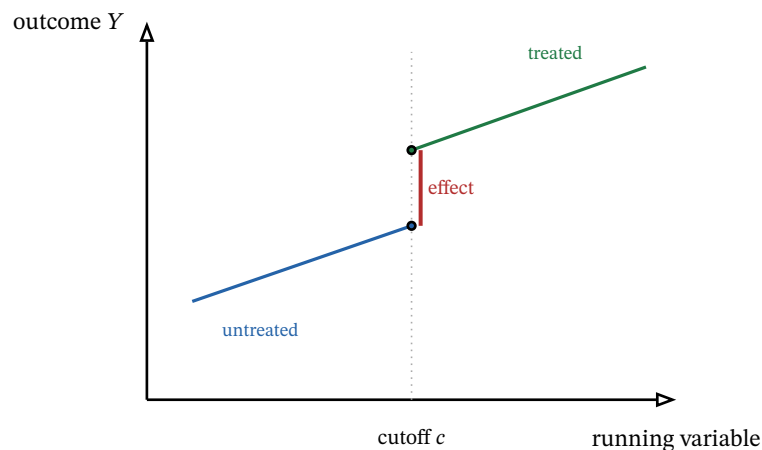


Figure 10: Regression discontinuity. Units are ordered by a **running variable**; crossing the **cutoff** c switches treatment on. Because units just below and just above c are near-identical, the **jump** in the outcome at c — not the slope on either side — is the causal effect. Maimonides’ rule supplies such a cutoff: class size drops sharply as enrolment crosses a multiple of 40.

9.5. Difference-in-differences: generalising the comparison

The natural-experiment comparison becomes a general-purpose tool once we allow **any** treated group, **any** control group, and a counterfactual built from their common trend. That tool is difference-in-differences.

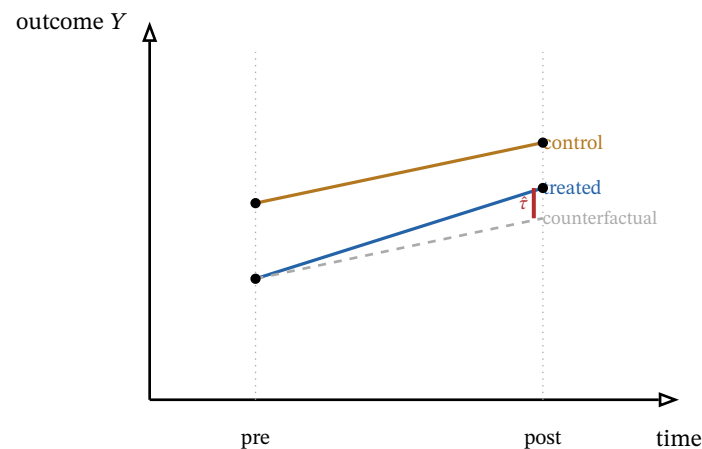


Figure 11: Difference-in-differences. The **control** group’s pre-to-post change estimates the common trend. Applying that trend to the treated group’s starting point gives the dashed **counterfactual**. The treatment effect $\hat{\tau}$ is the gap between the treated group’s **observed** post outcome and that counterfactual — the difference of the two before–after changes.

Definition 28. With a treated group and a control group observed before and after a policy, the **difference-in-differences** estimator is the difference of the two before–after changes:

$$\hat{\tau} = (\bar{Y}_{T,\text{post}} - \bar{Y}_{T,\text{pre}}) - (\bar{Y}_{C,\text{post}} - \bar{Y}_{C,\text{pre}}).$$

It nets out fixed differences between the groups **and** any common time trend. Its identifying assumption is **parallel trends**: absent the policy, treated and control would have moved together.

Example — Card–Krueger as difference-in-differences. New Jersey is the treated group, Pennsylvania the control. Suppose average full-time-equivalent employment per restaurant moved as follows:

$$\text{NJ: } 20.4 \rightarrow 21.0 \quad (+0.6), \quad \text{PA: } 23.3 \rightarrow 21.2 \quad (-2.1).$$

Then $\hat{\tau} = (+0.6) - (-2.1) = +2.7$ FTE per restaurant. The before/after change **within** New Jersey (+0.6) would have understated things; differencing out Pennsylvania’s common downturn (−2.1) isolates the policy’s effect — and it is not negative. (These stylised figures reproduce the sign and rough magnitude of Card and Krueger’s finding.)

Pitfall. A causal claim is only as good as its **named identifying assumption**. For DiD that is parallel trends — and it **fails** if something else hit the two groups differently at the same time, or if the groups were already diverging beforehand (check pre-trends!). “Treated and control just differ by X ” is not identification until you have argued the counterfactual is credible. Correlation plus a fitted line is not a causal effect.

Exercises

9.1. State the **fundamental problem of causal inference** in one sentence, and say why a raw before-and-after comparison is not enough.

9.2. A region rolls out a training programme. Average monthly earnings in the treated region move €1 800 → €2 100; in an untreated control region they move €1 750 → €1 900. (a) Compute the difference-in-differences estimate. (b) Name its identifying assumption. (c) Give one way that assumption could fail, and how you would check for it.

9.3. For each design, state the identifying assumption it needs and one threat to it. (a) An RD study grants a hiring subsidy only to firms with at most 50 employees, and compares firms just under and just over 50. (b) An IV study uses distance to the nearest college as an instrument for years of schooling. (c) In one sentence, say why a randomised controlled trial is the **benchmark**, and give one reason it may still fail to answer a labour-market policy question.

Answers. (1) A unit’s outcomes **with** and **without** treatment can never both be observed, so any raw comparison confounds the policy with every other difference between the groups or periods.

(2) $\hat{\tau} = (2100 - 1800) - (1900 - 1750) = 300 - 150 = +€150$ per month. Identifying assumption: **parallel trends** — absent the programme, treated and control regions would have moved together. It fails if a different shock hit one region (a local boom or plant closure) or if the two were already diverging; check by comparing their **pre-programme** earnings trends.

(3) (a) RD assumes firms just below and just above 50 are otherwise identical, so the jump in the outcome at 50 is the subsidy’s effect; the threat is **manipulation** — firms staying at 49 to keep the subsidy, which makes those just below unlike those just above. (b) IV needs **relevance** (distance shifts schooling) and **exclusion** (distance affects earnings only through schooling); the threat is that living far from a college also proxies for rural labour markets or family background, a second path to earnings. (c) An RCT is the benchmark because random assignment makes treated and control identical in expectation, building the counterfactual rather than assuming it; but many policies cannot be randomised, and a trial’s result may not generalise beyond the sample it ran on (external validity).

10. Automation and labour-market polarisation

We close on the open question that worries many students entering the job market: will automation take the jobs? Here the discipline of the previous chapter matters most, because the topic is full of confident claims the evidence does not support in either direction.

10.1. Routine versus non-routine tasks

Example — two jobs, two fates. A payroll clerk and a home-care nurse. The clerk's core tasks are **routine**: codifiable, rule-based, exactly what software does cheaply — so that occupation has shrunk. The nurse's tasks are **non-routine**: physical dexterity, judgement, and human contact resist codification — so that occupation has grown. The dividing line that predicts exposure is not “high-skill versus low-skill” but **routine versus non-routine**.

Definition 29. The **task-based view** classifies work by whether its tasks are **routine** (codifiable, rule-based) or **non-routine**, and whether they are **cognitive** or **manual**. Technology **substitutes** most readily for routine tasks (which can be programmed) and **complements** non-routine tasks (raising their productivity and the demand for them).

10.2. Polarisation and complementarity

Definition 30. **Job polarisation** is the empirical hollowing-out of **middle-skill, routine** occupations relative to **high-skill non-routine cognitive** jobs and **low-skill non-routine manual** jobs — a U-shaped change in employment across the skill distribution, documented across the United States and Europe since roughly 1980. Where machines **complement** labour, automation raises wages and employment in the affected tasks; where they **substitute**, it displaces workers. This complement-versus-substitute split is the elasticity of substitution σ from Chapter 2, now applied task by task.

Pitfall. Be scrupulous about **what economics can and cannot adjudicate** here:

- **Can:** that routine tasks are the most exposed to automation, and that the past four decades show measurable polarisation — these are well-identified empirical findings.
- **Cannot (yet):** the **net long-run effect on total employment and wages**, because automation simultaneously **destroys** tasks and **creates** new ones, and the pace and reach of future technology — including AI moving into non-routine **cognitive** work, historically the least exposed — are genuinely unknown. A forecast of mass technological unemployment and a forecast of seamless reabsorption are **both** beyond what current evidence can settle. Honesty about that boundary — saying plainly what the evidence cannot yet decide — is itself part of reasoning like an economist.

10.3. Policy responses: active labour-market policies and flexicurity

Whatever the net long-run verdict, displacement is real for the individual worker, and how a country cushions and redirects it is a policy choice with measurable consequences. Two ideas organise the modern toolkit.

Example — two ways to help a laid-off worker. A plant closes and a machine operator loses her job. One country simply mails her an unemployment cheque — a **passive** transfer that protects her income while she searches. Another adds a caseworker, pays for a retraining course in a growing occupation, and subsidises the first months of a new employer's wage bill — **active** measures aimed at getting her back into work, and into a better-matched job. Most systems do both; the balance between them is the policy lever.

Definition 31. Active labour-market policies (ALMPs) promote employment directly rather than merely replacing lost income. The main instruments are

- **job-search assistance:** counselling, matching help, and monitoring of search;
- **training and retraining:** rebuilding the human capital of Chapter 3 when old skills lose their market;
- **hiring subsidies** and direct public job creation.

They stand opposite **passive** policies — unemployment benefits — which insure income but neither create nor find jobs. When ALMPs are evaluated with the natural-experiment and difference-in-differences methods of Chapter 9, job-search assistance and well-targeted private-sector wage subsidies tend to be the most cost-effective; pure public-job-creation schemes fare worst; and the returns to training are real but slow, showing up mainly in the medium run.

Definition 32. Flexicurity combines three ingredients a naïve reading of the competitive model treats as at odds:

1. **flexibility** — light hiring-and-firing rules, so firms adjust employment freely;
2. **security** — generous unemployment insurance, so a job loss is cushioned (income security, not job security);
3. **activation** — strong ALMPs that move the insured worker quickly back into work.

Denmark is the standard case (the flexicurity row of the comparative table in Chapter 6): low job protection, high benefits, heavy activation spending. The bet is that security and activation together let the labour market reallocate workers fast **without** leaving them exposed.

Pitfall. Recall from Chapter 6 that generous benefits raise the **reservation wage** and lengthen search — taken alone they should **raise** frictional unemployment. The flexicurity claim is not that this effect vanishes but that **activation offsets it:** monitoring and job-search assistance pull the reservation wage and search duration back down. Copy the “security” leg without the “activation” leg and you get the long-search problem without the cure — the components are a package, not a menu.

Exercises

10.1. Using the routine/non-routine task framework, explain job polarisation and the complement/substitute distinction (relating it to the elasticity of substitution). Then list precisely which claims about automation's effect on employment current economics can adjudicate and which remain genuinely uncertain, and why.

10.2. (a) Distinguish **active** from **passive** labour-market policies and give two examples of each. (b) State the three components of **flexicurity** and explain why, despite generous benefits raising the reservation wage, the Danish model need not produce high unemployment. (c) A government copies flexicurity's generous benefits but funds no activation. Predict the effect on search duration and frictional unemployment, and justify it.

Answers. **(1)** Routine tasks (codifiable) are substituted by machines; non-routine tasks are complemented — a task-level reading of σ . Polarisation is the U-shaped hollowing of middle-skill routine jobs. Economics **can** adjudicate that routine work is most exposed and that measurable polarisation occurred since 1980; it **cannot yet** settle the net long-run employment/wage effect, because destruction and creation of tasks run together and future technology (including AI in cognitive work) is genuinely uncertain.

(2) Active policies promote employment directly (job-search assistance, training, hiring subsidies); passive policies replace income (unemployment benefits). The three flexicurity components are flexibility (light hire/fire rules), security (generous benefits), and activation (strong ALMPs); benefits alone raise the reservation wage and lengthen search, but activation offsets this by monitoring search and speeding re-employment, so the package need not raise unemployment. Copying the benefits without activation keeps the reservation-wage rise uncured: search duration and frictional unemployment go up.

Primary sources

Each pointer is the original work behind a result used above; the chapter it serves is named in brackets.

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Further reading

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